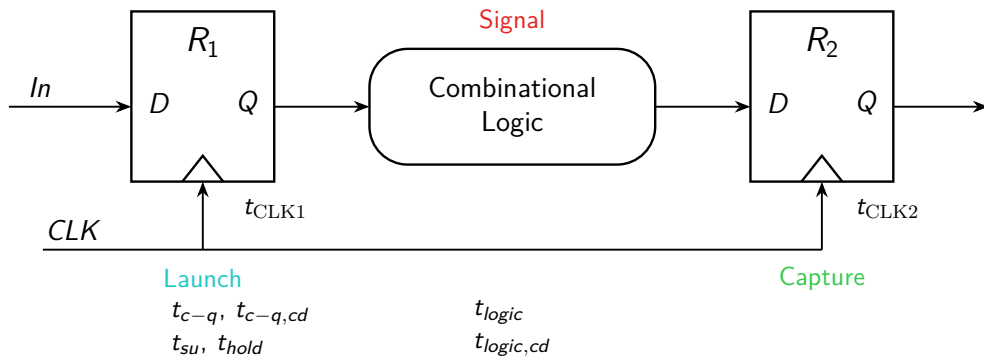
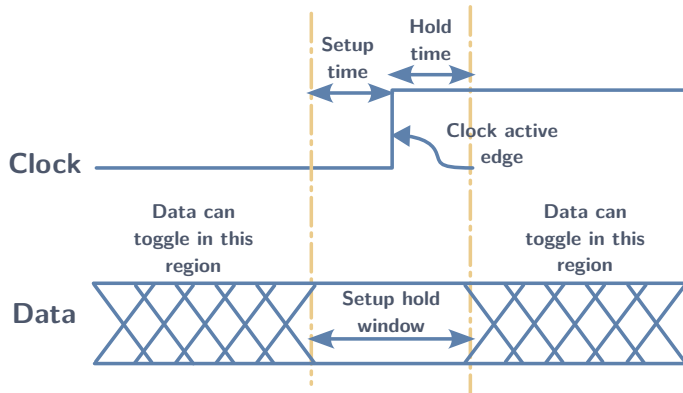


Register-to-register timing path



Setup and hold window



Define the timing contract

```
# Define a 0.510 ns core clock.
create_clock -name core_clock -period 0.510 \
  [get_ports clk]

# Reserve 0.102 ns at each input.
# Exclude the clock port itself.
set_input_delay 0.102 -clock core_clock \
  [lsearch -inline -all -not -exact \
    [all_inputs] [get_ports clk]]

# Reserve 0.102 ns at each output.
set_output_delay 0.102 -clock core_clock \
  [all_outputs]
```

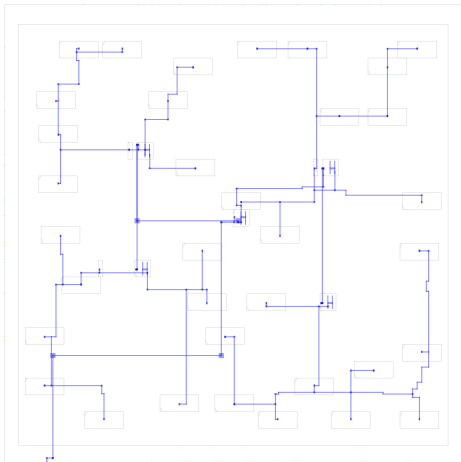
Square brackets in timing Tcl

```
create_clock -period 0.510 [get_ports clk]
set_output_delay 0.102 -clock core_clock \
    [all_outputs]
```

Tcl runs the command inside each pair of square brackets first. `[get_ports clk]` supplies the clock port to `create_clock`; `[all_outputs]` supplies the output ports to `set_output_delay`.

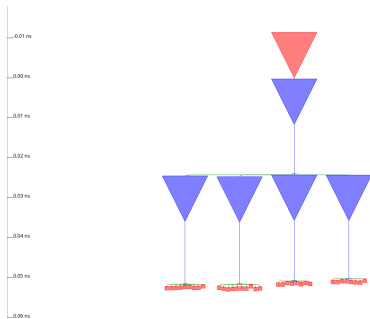
In command references, square brackets around an option mean it is optional. In a Tcl script, square brackets execute a command.

Clock-tree synthesis and timing repair



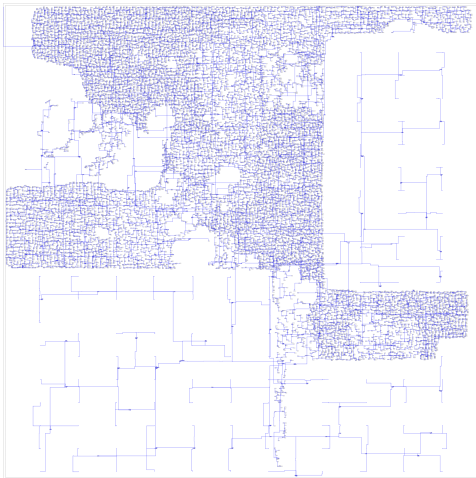
```
# Normalize pre-existing clock inverters so
CTS sees a consistent tree.
repair_clock_inverters
clock_tree_synthesis -root_buf BUF_X4 \
  -buf_list BUF_X4 -sink_clustering_enable \
  -sink_clustering_max_diameter 100
repair_clock_nets
# Legalize CTS cells; timing repair may move
or insert cells, so
# detailed placement is repeated after
repair_timing.
detailed_placement
set_propagated_clock [all_clocks]
estimate_parasitics -placement
repair_timing -skip_gate_cloning
detailed_placement
```

Clock skew and slack reports



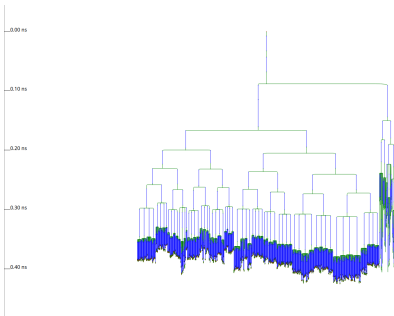
```
# Measure clock arrival mismatch and the  
# worst setup/hold margins.  
# The -max report is setup; the -min report  
# is hold.  
report_clock_skew -digits 3  
report_worst_slack -max -digits 3  
report_worst_slack -min -digits 3
```

Show the physical clock nets



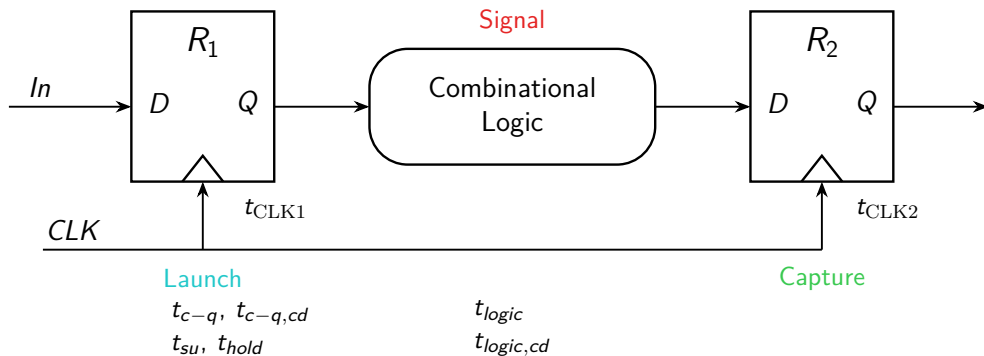
```
# Hide the module overlay and ordinary nets;  
show only the clock  
# network, clock-tree cells, and sequential  
endpoints.  
gui::set_display_controls "Misc/Module view"  
visible false  
gui::set_display_controls "Nets/*" visible  
false  
gui::set_display_controls "Nets/Clock"  
visible true  
gui::set_display_controls \  
  "Instances/StdCells/Clock tree/*" visible  
true  
gui::set_display_controls \  
  "Instances/StdCells/Sequential" visible  
true
```

Open the Clock Tree Viewer

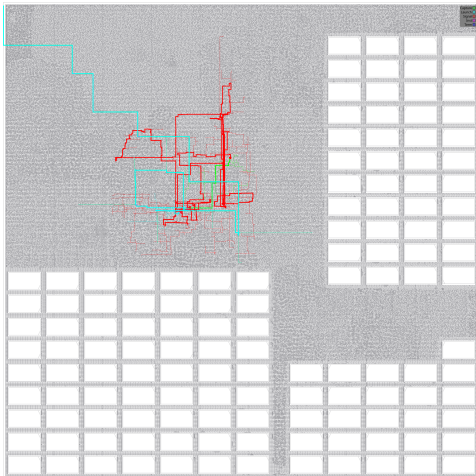


```
# CTS must have created a clock tree, and the  
    final DB must be loaded  
# with the clock definition plus the timing  
    libraries/constraints.  
# Open the widget, select a clock, then use  
    Update and Fit in the GUI.  
gui::show_widget "Clock Tree Viewer"  
gui::select_clockviewer_clock core_clock
```

Register-to-register timing path



Read a selected timing path



```
# Draw the worst setup path in the layout.  
Clicking a Timing Report  
# row selects that row's path; report_checks  
prints the path in text.  
# Launch: source register. Capture:  
destination register.  
# Signal: data path. Inst: cell. Term: pin or  
terminal.  
gui::show_worst_path -setup  
report_checks -path_delay max \\  
    -format full_clock_expanded \\  
    -fields {input_pin slew capacitance}  
-digits 3
```

Read the Timing Report table

File View Tools Windows Options Help

Fit Find Inspect Timing

Display Control

Layers

- Other
 - metal1
 - via1
 - metal2
 - via2
 - metal3
 - via3
 - metal4
 - via4
 - metal5
 - via5
 - metal6
 - via6
 - metal7
 - via7
 - metal8
 - via8
 - metal9
 - via9
 - metal10
- Nets
- Instances
- Blockages
- Rulers
- Rows
- Tracks
- Shape Types
- Misc
- Heat Maps
- Timing Path

Timing Report

Settings Columns Update

Setup / Hold

Capture Clock	Required	Arrival	Slack	Skew	Logic Del.
core_clock	10.350	6.460	3.900	-0.018	5.
core_clock	10.365	6.463	3.902	-0.024	5.
core_clock	10.365	6.463	3.902	-0.024	5.
core_clock	10.364	6.462	3.903	-0.023	5.
core_clock	10.358	6.455	3.903	-0.016	5.
core_clock	10.364	6.461	3.903	-0.023	5.
core_clock	10.367	6.463	3.905	-0.026	5.
core_clock	10.367	6.462	3.905	-0.026	5.
core_clock	10.362	6.456	3.906	-0.021	5.
core_clock	10.367	6.460	3.907	-0.026	5.
core_clock	10.340	6.431	3.908	0.001	5.
core_clock	10.366	6.456	3.910	-0.024	5.

Data Path Details

Pin	Fanout	tl	Time	Delay	Skew	Load
clk_i	1	↑	0.000	0.000	0.000	33.512
clock network delay			0.380	0.380		
issue_stage_i/i_scoreboard/_28029/_CK (DFFR_X1)		↑	0.380	0.001	0.010	
issue_stage_i/i_scoreboard/_28029/_Q (DFFR_X1)	5	↑	0.513	0.133	0.036	14.577
issue_stage_i/i_scoreboard/_28026/_A (HA_X1)		↑	0.513	0.000	0.036	
issue_stage_i/i_scoreboard/_28026/_CO (HA_X1)	2	↑	0.564	0.052	0.019	6.680
issue_stage_i/i_scoreboard/_28028/_A (HA_X1)		↑	0.565	0.000	0.019	
issue_stage_i/i_scoreboard/_28028/_CO (HA_X1)	11	↑	0.694	0.129	0.096	42.922

Timing Report Help Browser

Read the slack chart

